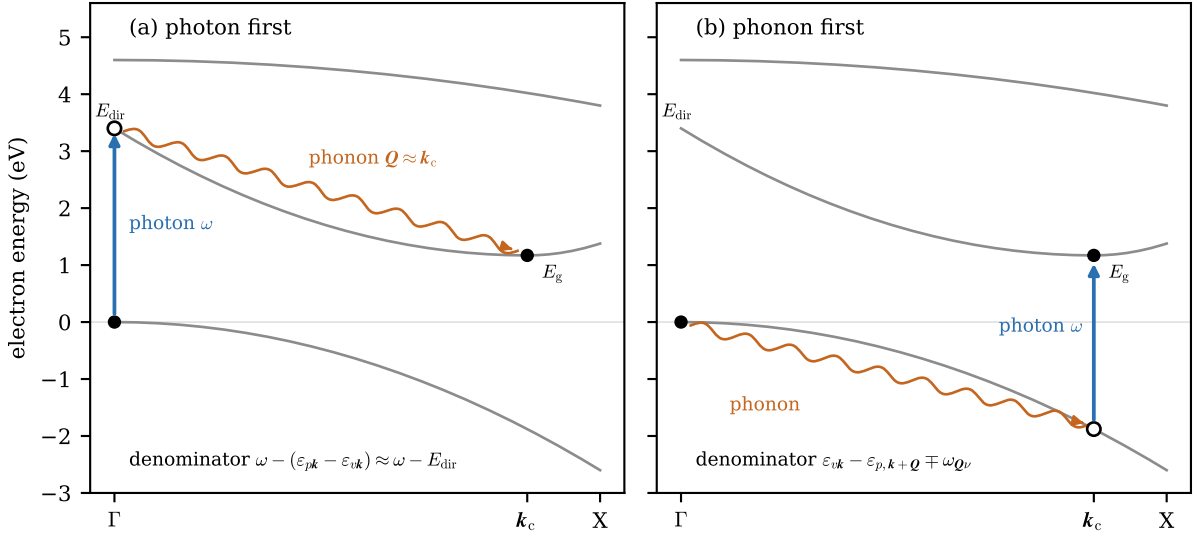
(a) photon first
(b) phonon first
electron energy (eV)
5
4
3
2
1
0
−1
−2
−3
E_{dir}
E_{g}
phonon $\boldsymbol{Q} \approx \boldsymbol{k}_{\mathrm{c}}$
photon ω
phonon
denominator $\omega - (\varepsilon_{p\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}}) \approx \omega - E_{\mathrm{dir}}$
denominator $\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}$
Γ
$\boldsymbol{k}_{\mathrm{c}}$
X